

1. Give an example how each of the following may be used on the table that you have created:

1) ALTER TABLE

It is use to add an new column to employees

```
ALTER TABLE employees ADD COLUMN partners VARCHAR(50);
```

2) DROP TABLE

Deletes the students table

```
DROP TABLE employees;
```

3) RENAME TABLE

It is use to rename a table

```
RENAME TABLE employees TO personnel;
```

4) TRUNCAT

Delete all data in the table, but keep the table

```
TRUNCATE TABLE employees;
```

2. What is the distinction between TRUNCATE, DELETE, and DROP for tables?

DELETE removes specific rows from a table using WHERE. It can be rolled back if inside a transaction. The table and its structure stay the same.

TRUNCATE deletes all rows in a table quickly. It can't use WHERE, and usually can't be rolled back. The table stays, but it's empty.

DROP completely removes the entire table, including its structure and data. This action is permanent and cannot be undone.

5. Add the three new employees to the o_employees table– using the syntax:

```
INSERT INTO o_employees (employee_id, first_name, last_name, email,  
hire_date, job_id) VALUES(210, 'Ramon', 'Sanchez', 'RSANCHEZ', SYSDATE,
```

```
'HR_MAN');
```

```
INSERT INTO o_employees (employee_id, first_name, last_name, email,  
hire_date, job_id) VALUES(211, 'Alba', 'Varro', 'AVARRO', SYSDATE,  
'HR_MAN');
```

```
INSERT INTO o_employees (employee_id, first_name, last_name, email,  
hire_date, job_id) VALUES(212, 'Fernando', 'Sam', 'FSAM', SYSDATE,  
'HR_MAN');
```

7. CREATE a table called Artists.
a Add the following to the table:

```
CREATE TABLE artists (  
  artist_id INT PRIMARY KEY,  
  first_name VARCHAR(50),  
  last_name VARCHAR(50),  
  band_name VARCHAR(100),  
  email VARCHAR(100),  
  hourly_rate DECIMAL(10, 2)  
);
```

b INSERT one artist from the d_songs table.

```
INSERT INTO artists (artist_id, first_name, last_name, band_name, email, hourly_rate)  
VALUES (45, NULL, NULL, 'The Hobbits', 'the@hobbits.com', 50.00);
```

c INSERT one artist of your own choosing.

```
INSERT INTO artists (artist_id, first_name, last_name, band_name, email, hourly_rate)  
VALUES (55, 'Calogero', NULL, 'Calogero', 'the@calogero.com', 55.00);
```

8. SELECT last_name || ' ' || first_name AS "Name"
FROM o_employees;

9. SELECT last_name || ' ' || first_name AS "Name"
FROM o_employees
WHERE manager_id=124;

10. SELECT last_name || ' ' || first_name AS "Name"
FROM o_employees
WHERE salary BETWEEN 7000 AND 10000;

11. SELECT last_name || ' ' || first_name AS "Name"
FROM o_employees
WHERE job_id ='AD_VP';

12. SELECT min_salary, max_salary
FROM o_jobs
WHERE job_id ='AD_VP';

13. SELECT last_name, first_name, salary, salary+salary*0.15
FROM o_employees
WHERE last_name like 'G%';

14. SELECT last_name, first_name, salary, salary+salary*0.05 as "New salary"
FROM o_employees
WHERE manager_id=124 and salary>3000;

15. SELECT *
FROM o_employees NATURAL JOIN o_departments ;

16. SELECT first_name, last_name, department_name
FROM o_employees JOIN o_departments USING (department_id)
WHERE department_id=80;

17. SELECT *
FROM o_employees JOIN o_jobs USING (job_id);

18. SELECT first_name, last_name, min_salary, max_salary
FROM o_employees JOIN o_jobs USING (job_id)
WHERE job_id='AD_VP';

19. SELECT *
FROM o_employees NATURAL JOIN o_jobs NATURAL JOIN o_departments;

20. SELECT first_name, last_name, max_salary, department_name, location_id
FROM o_employees NATURAL JOIN o_jobs NATURAL JOIN o_departments
WHERE salary > 9000;

21. SELECT first_name, last_name, max_salary, department_name, location_id
FROM o_employees NATURAL JOIN o_jobs NATURAL JOIN o_departments
WHERE hire_date < '01-Jan-1999';

22. SELECT first_name, last_name, department_name
FROM o_employees CROSS JOIN o_departments;

23. SELECT *
FROM o_employees NATURAL JOIN o_departments;

24. SELECT department_id, department_name, first_name, last_name, salary
FROM o_employees NATURAL JOIN o_departments
WHERE department_name = 'IT' ;

25. SELECT *
FROM o_employees JOIN o_departments USING (manager_id);

26. SELECT department_id, department_name, first_name, last_name
FROM o_employees NATURAL JOIN o_departments
WHERE department_id IN (80, 90);

27. SELECT employee_id, department_name, hire_date, last_name
FROM o_employees NATURAL JOIN o_departments
WHERE hire_date = '07-Jun-1994';

28. SELECT first_name, last_name, salary, hire_date, department_name, location_id
FROM o_employees NATURAL JOIN o_departments;

29. SELECT first_name, last_name, salary, hire_date, department_name, manager_id
FROM o_employees NATURAL JOIN o_departments
WHERE department_id=80;

30. SELECT salary, bonus, first_name, last_name, department_name, manager_id
FROM o_employees NATURAL JOIN o_departments;

31. SELECT salary, bonus, first_name, last_name, department_name, manager_id
FROM o_employees NATURAL JOIN o_departments
WHERE manager_id=124;

32. SELECT first_name, last_name, location_id
FROM o_employees JOIN o_departments USING (manager_id);

33. CREATE TABLE copy_employees AS (SELECT * FROM employees);

34. CREATE TABLE copy_departments AS (SELECT * FROM departments);

35.

a. UPDATE copy_employees
SET salary = 1200
WHERE last_name = 'Abel'

b. UPDATE copy_employees
SET email = 'ssking'
WHERE last_name = 'King'

c. UPDATE copy_employees
SET phone_number = 5151234570
WHERE first_name = 'Lex'

d. UPDATE copy_employees
SET job_id = 'ST_Clerk1'
WHERE department_id=50

e. UPDATE copy_employees
SET salary = 5000
WHERE department_id=50

f. UPDATE copy_employees
SET salary = (SELECT salary FROM copy_employees WHERE employee_id = 100)
WHERE employee_id=101

g. UPDATE copy_employees
SET hire_date = (SELECT hire_date FROM copy_employees WHERE last_name = 'Matos')
WHERE last_name = 'Grant';

h. UPDATE copy_employees
SET department_id = (SELECT department_id FROM copy_employees WHERE
first_name = 'Taylor')
WHERE last_name = 'Mourgos';

i. UPDATE copy_employees
SET department_id = 70
WHERE manager_id=100;

j. UPDATE copy_employees
SET salary = 9500
WHERE job_id= 'IT_PROG';